STATISTICS & PROBABILITY
CHAPTER-I

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Concept of data and dataset

❖**What is Data?**

- ✓ **Data** means **individual pieces of information**.
- ✓ Think of data as **single facts** or **values**.
- ✓ Example:
  - ✓ A person's age: **25**
  - ✓ A house price: **$200,000**
  - ✓ A student's grade: **A**

Concept of data and dataset

❖**What is a Dataset?**

- ✓ A **dataset** is a **collection of data**, usually organized in a table.
- ✓ In ML and data science, we use datasets to **train, validate, and test models**.
- ✓ Example: A table of house prices with multiple columns:

Size (sqft)	Bedrooms	Location	Price
1200	3	City A	$200,000
1500	4	City B	$250,000
900	2	City C	$150,000

Here:

- Each **row** = one example (or one data point), like one house.
- Each **column** = a feature or attribute (size, bedrooms, etc.).

Concept of data and dataset

CGPA	Offered Job?
2.4	No
2.8	No
3.1	Yes
3.5	Yes
2.6	No
3.7	Yes
3.0	Yes
2.5	No
3.3	Yes
2.9	No
3.6	Yes
3.8	Yes

Here we are classifying students whether a student/candidate getting job or not. Hence, it is classification problem of machine learning.

Concept of data and dataset

CGPA	Salary Offered (in K)
3.2	45
3.5	50
3.8	60
2.9	40
3.7	58
3.0	42
3.9	65
3.4	48
3.1	44
3.6	55

Here we are classifying students whether a student/candidate getting job or not. Hence, it is classification problem of machine learning.

Concept of ML

- **Machine learning** is a field of artificial intelligence (AI) that teaches computers to **learn from data** and **make predictions or decisions without being explicitly programmed**.

- How is it different from traditional programming?

Traditional programming:

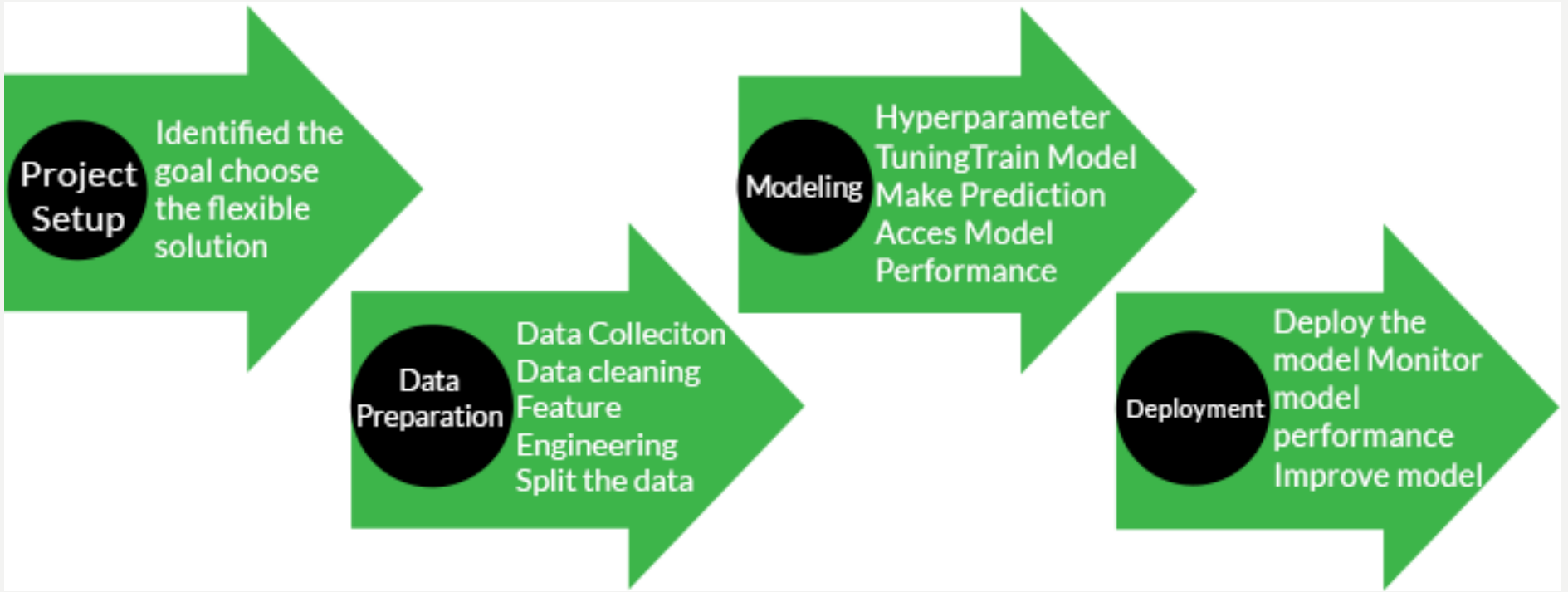
You write rules (logic) + give data ➜ get output.

Machine learning:

You give data + output (examples) ➜ the machine learns rules (patterns).

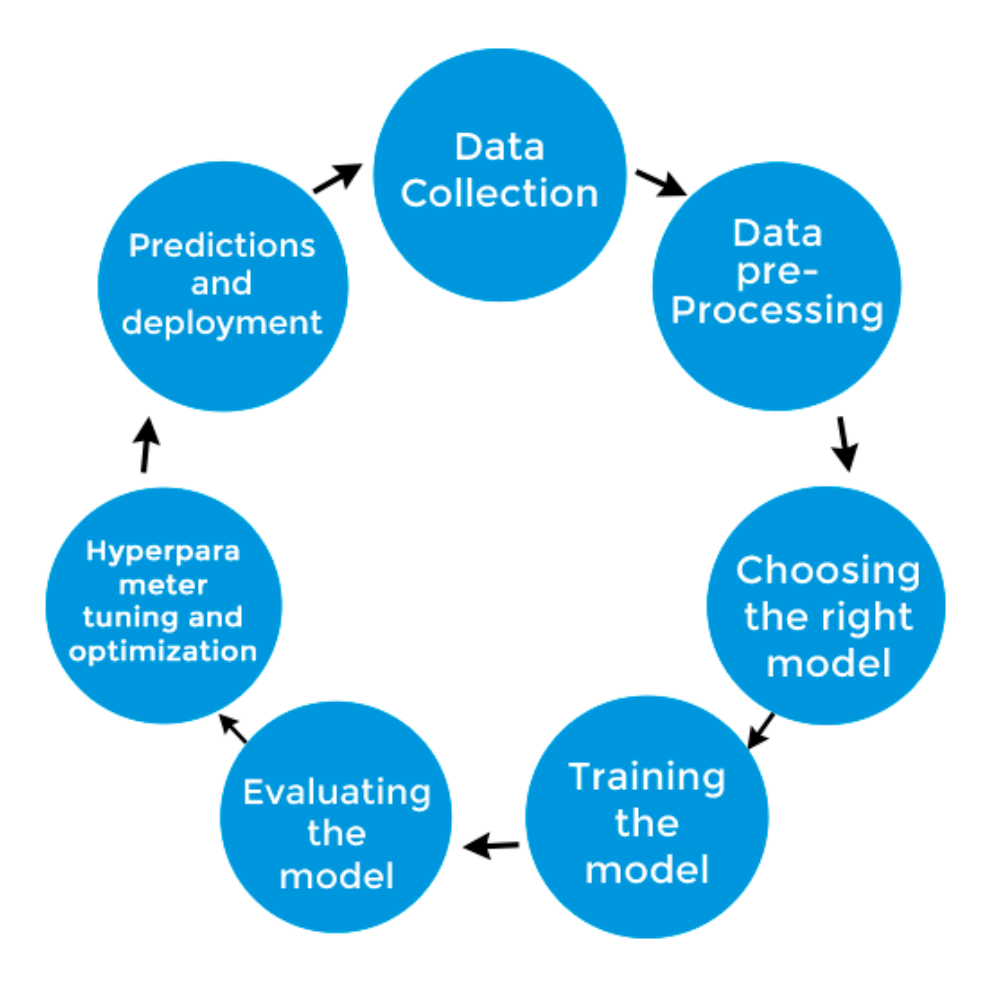
How does Machine Learning Work?

- Machine Learning process includes Project Setup, Data Preparation, Modeling and Deployment. The following figure demonstrates the common working process of Machine Learning. It follows some set of steps to do the task; a sequential process of its workflow is as follows:



Stages of Machine Learning

❖The following are the stages (detailed sequential process) of Machine Learning:



Stages of Machine Learning

- **Data Collection** – Data collection is an initial step in the process of machine learning. In this stage, it collects data from the different sources such as databases, text files, pictures, sound files, or web scraping. This process organizes the data in an appropriate format, such as a CSV file or database, and makes sure that they are useful for solving your problem.

- **Data Pre-processing** – It is a key step in the process of machine learning, which involves deleting duplicate data, fixing errors, managing missing data either by eliminating or filling it in, and adjusting and formatting the data.

- **Choosing the Right Model** – The next step is to select a machine learning model; once data is prepared, then we apply it to ML models like linear regression, decision trees, and neural networks that may be selected to implement. This selection depends on many factors, such as the kind of data and your problem, the size and type of data, the complexity, and the computational resources.

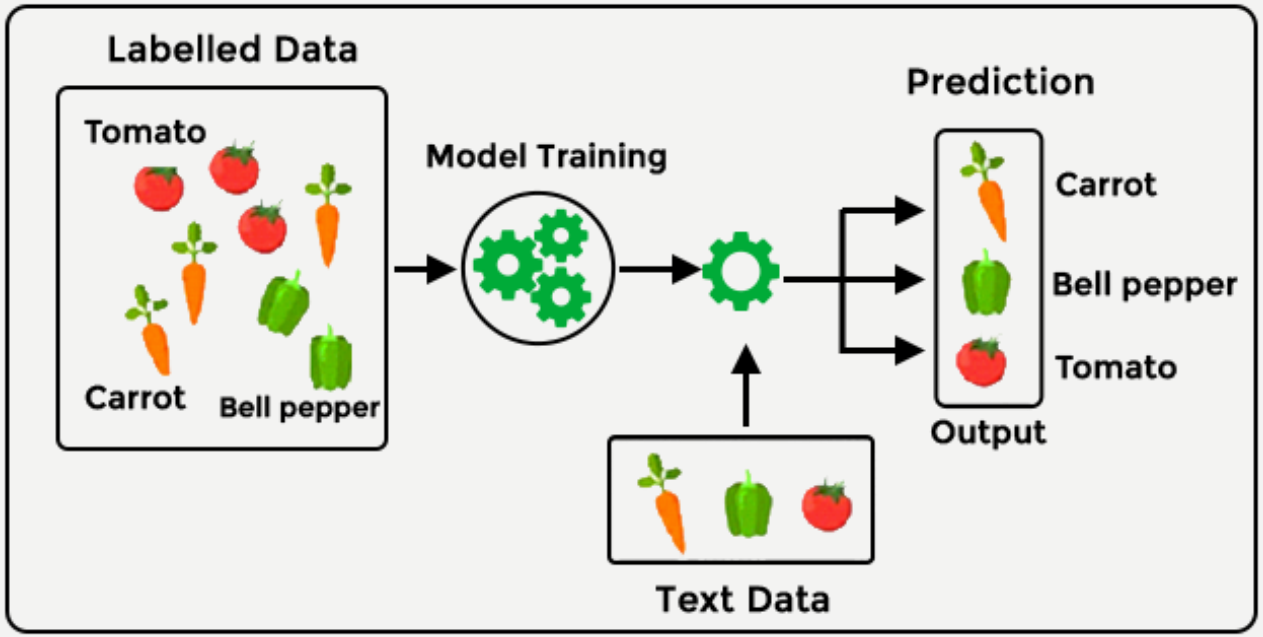
Stages of Machine Learning

- **Training the Model** – This step includes training the model from the data so it can make better predictions.
- **Evaluating the model** – When module is trained, the model has to be tested on new data that they haven't been able to see during training.
- **Hyperparameter Tuning and Optimization** – After evaluating the model, you may need to adjust its hyperparameters to make it more efficient. You should try different combinations of parameters and cross-validation to ensure that the model performs well on different data sets.
- **Predictions and Deployment** – When the model has been programmed and optimized, it will be ready to estimate new data. This is done by adding new data to the model and using its output for decision-making or other analysis. The deployment includes its integration into a production environment to make it capable of processing real-world data.

Types of Machine Learning

❖Machine learning models fall into the following categories:

1) **Supervised Machine Learning** – It is a type of machine learning that trains the model using labeled datasets to predict outcomes.



Supervised learning algorithms are generally categorized into two main types:

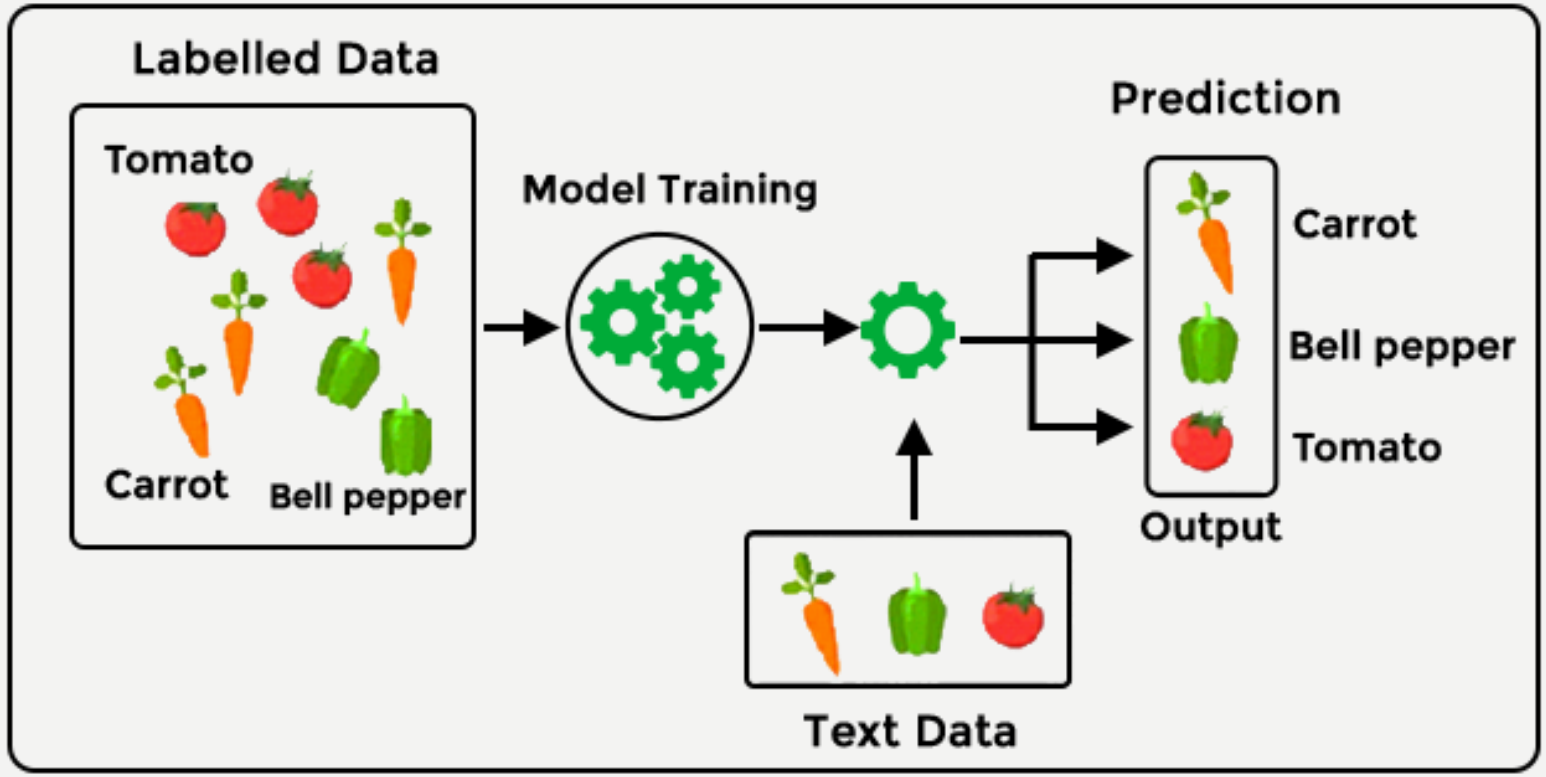
Classification - where the goal is to predict discrete labels or categories.

Regression - where the aim is to predict continuous numerical values.

Types of Machine Learning

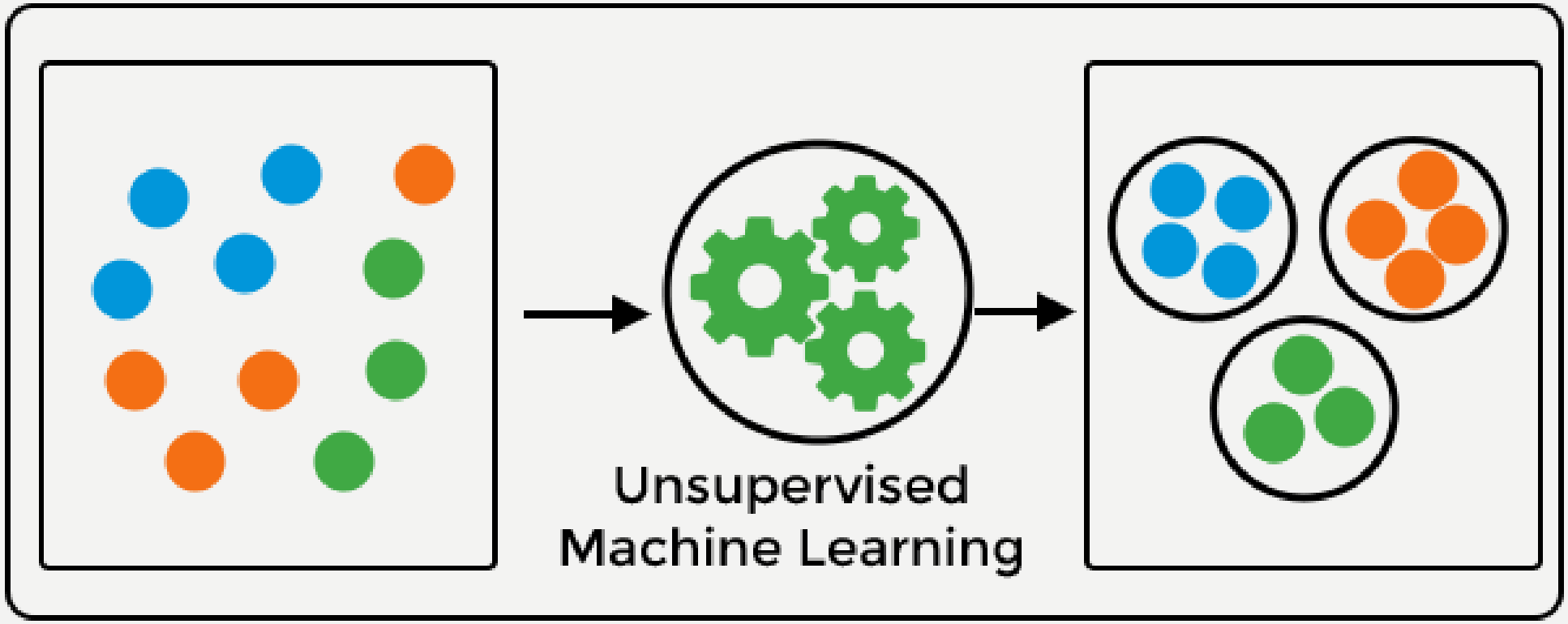
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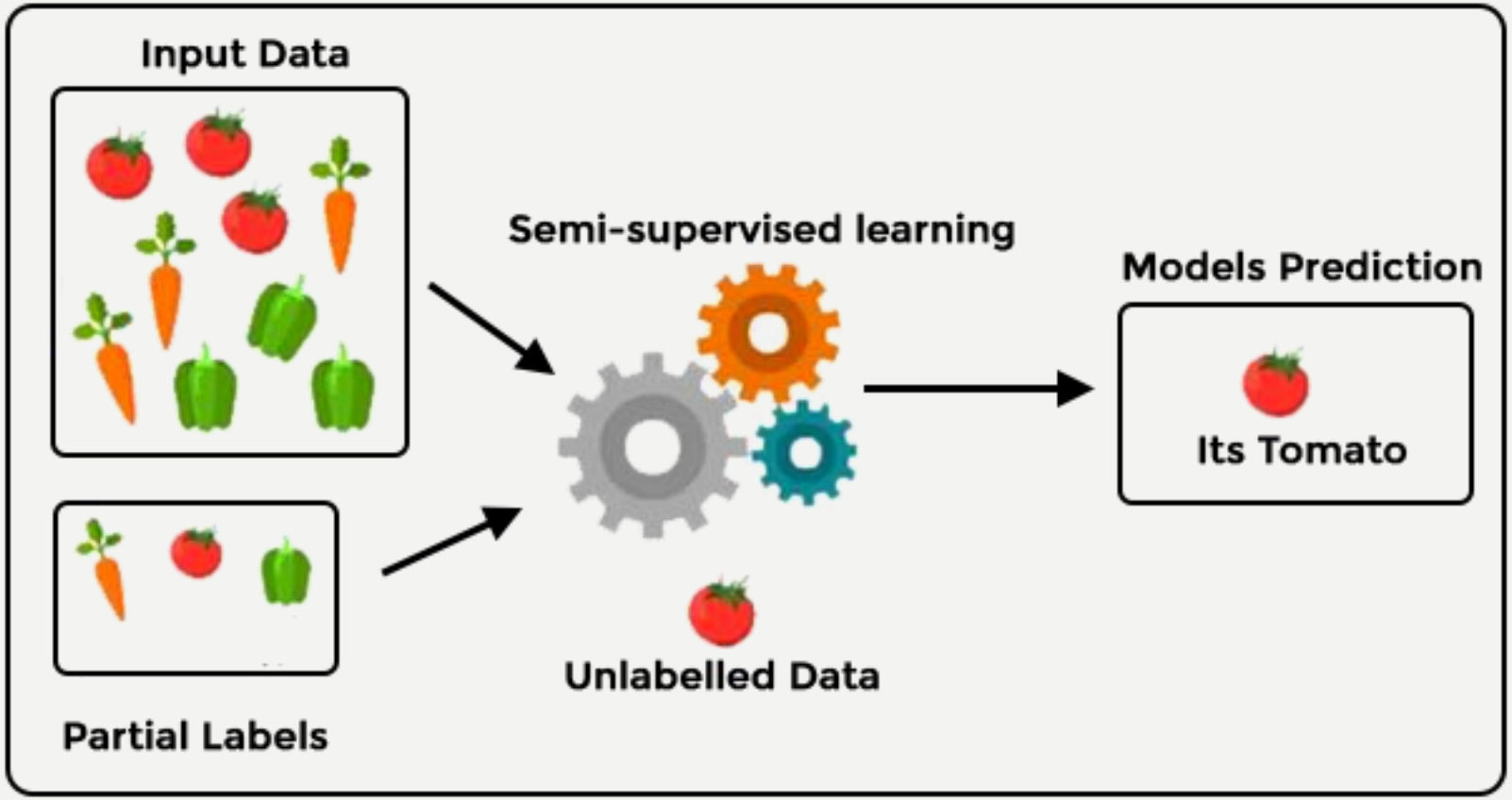
Types of Machine Learning

2) Unsupervised Machine Learning – It is a type of machine learning that learns patterns and structures within the data without human supervision.



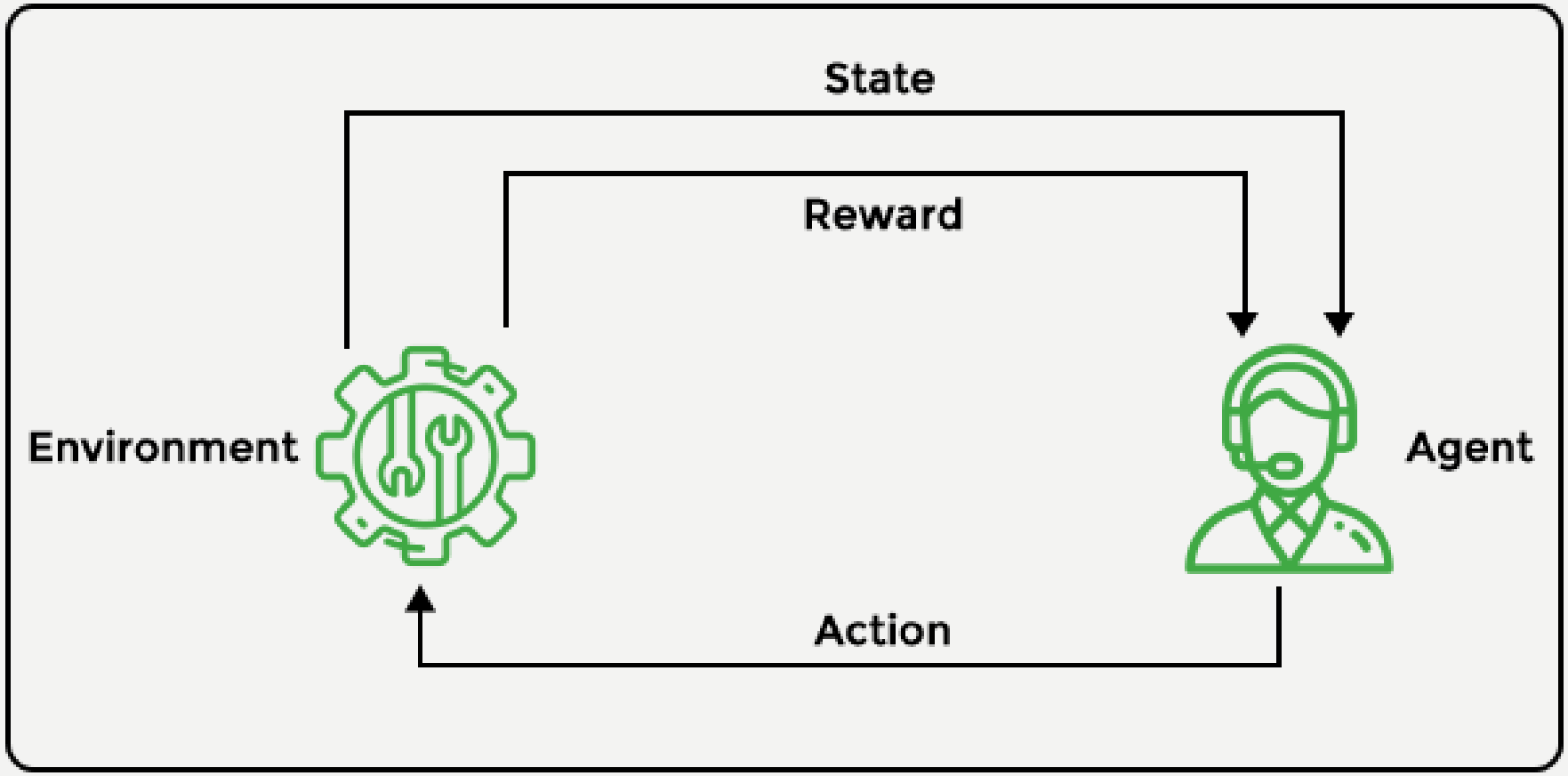
Types of Machine Learning

3) **Semi-supervised Learning** – It is a type of machine learning that is neither fully supervised nor fully unsupervised. The semi-supervised learning algorithms basically fall between supervised and unsupervised learning methods.



Types of Machine Learning

4) **Reinforcement Machine Learning** – It is a type of machine learning model that is similar to supervised learning but does not use sample data to train the algorithm. This model learns by trial and error.



Types of Machine Learning

Student Name	CGPA	Age	IQ Score	Height	Weight	Grade	Programming Skill	Offered Job
Alice	3.85	50	135	165	58	A	Advanced	Yes
Bob	3.20	—	115	172	70	B	Intermediate	No
Charlie	3.50	24	125	168	65	B	Advanced	Yes
Diana	2.80	25	105	158	55	C	Beginner	No
Ethan	3.75	23	128	180	75	A	Advanced	Yes
Fiona	3.60	22	130	162	60	A	Advanced	Yes
George	2.90	26	100	175	78	C	Beginner	No
Hannah	3.40	24	118	169	68	B	Intermediate	Yes
Ivan	3.10	23	110	177	72	B	Beginner	No
Julia	3.95	22	138	160	55	A	Advanced	Yes

Data Distribution

- A data distribution is a function F(X) that tells what possible values variable X can take and how many times each value x_i is present in the dataset or the frequency of x_i.

I'M EXCITED TO BE YOUR TEACHER!

- Share your name.
- I have been teaching _____ years.